

Scope

Self-prep filing for F-1 International students in their first 5 years for Washington

Principles

1. **Extract first, ask last (minimize input as much as possible)**
 - a. **EXTRACT** → Pulled silently from uploaded document
 - b. **CONFIRM** → Extracted, then single tap/click confirmation
 - c. **ASK** → Reserved for what cannot be extracted
 - d. **SYSTEM** → Automated Logic - No user involvement
2. **Split Brain (LLM explain, Engine calculate)**
 - a. Stage 0-4 and 6 → Agentic Layer (understanding, extraction, reasoning, explanation)
 - b. Stage 5 → Rules Engine (This is the actual Math)

Full Pipeline

Stage 0 - Residency Status

1. **Tax Year** - *Default to most recent tax year (disabled back filing for now)*
 - a. **ASK** → What tax year are you filing? (year)
 - b. **SYSTEM** → If not most recent year, DOES NOT SUPPORT
2. **Visa / Status type** - *Disabled any other status for now*
 - a. **EXTRACT** → Upload Visa in Passport Page
 - b. **CONFIRM** → Confirm status and make sure it's F1 (only support F1)
 - c. **SYSTEM** → If not F1, DOES NOT SUPPORT
3. **First Entry Date** - *More than five years of entry date means resident (does not support)*
 - a. **EXTRACT** → Upload I-94 travel history.
 - b. **CONFIRM** → Confirm their first date of entry
 - c. **ASK** → Were you ever in the US on an F, J, M, or Q visa in any earlier year? (Yes / No)
 - d. **SYSTEM** → If not less than 5 years, DOES NOT SUPPORT
4. **Green Card Check** - *Green card counted as resident*
 - a. **ASK** → Are you US lawful permanent resident (green card holder)?(Yes / No)
 - b. **SYSTEM** → If yes, DOES NOT SUPPORT

OUTPUT : Status Pass or No Pass with Reasoning (e.g. "Passed! You're a nonresident for within that window, so your days don't count toward residency")

EDGE CASE : Changing Visa Status mid-year. (NOT SUPPORTED for now)

Stage 1 - Document Intake

Determine Input and Output Document

1. **Working Status (W-2 FORM)**
 - a. **ASK** → Did you work in the US this year? (on-campus, CPT, OPT) [Yes/No]
 - i. If yes, **ASK** → Are you on OPT? (need EAD)
 - b. **SYSTEM** → If yes, need W-2 form

2. Social Security Number (SSN or ITIN or W-7)

a. **ASK** → Do you have Social Security Number (SSN) [Yes/No]

b. **SYSTEM**

i. **YES** → No need ITIN

ii. **NO** → Need ITIN

1. **ASK** → Have you ever had or applied for an ITIN? [Yes/No]

2. **SYSTEM**

a. **YES** → **ASK** → ITIN Number

b. **NO** → Need W-7 - **ITIN application process (not available for MVP)**

3. Scholarship / Fellowship (1042-S, 1098-T)

a. **ASK** → Did you receive scholarship or fellowship?

i. Yes, it covered tuition only

ii. Yes, it covered tuition & Living expenses?

iii. No

b. **SYSTEM** → If it covered living expenses, 1042-S and/or 1098t

4. Passive Income (1099-INT, 1099-DIV, 1099-B)

a. **ASK** → Did you earn any interest, dividends, or investment income in the US?

i. Bank Interest (Savings acc, CDs, High-Yields, savings) (1099-INT)

ii. Dividends (1099-DIV)

iii. Sold stocks, crypto, or other assets (any loss/profits) (1099-B)

iv. None of these

b. **SYSTEM** → If yes, then add those forms

OUTPUT → List of documents user needed to input and List of tax documents output

Stage 2 - Upload Files

1. Upload Field

a. **EXTRACT** → upload all listed documents (one drop zone) based on stage 1 Output

b. Documents Needed to upload

i. Passport (GENERAL)

ii. I-20 (GENERAL)

iii. EAD Card → Only for OPT

iv. **W-2 (Wage and Tax Statement)** → For any student doing CPT, OPT, or on-campus job

v. **1098-T** → For student with scholarship/fellowship

vi. **1042-S** → For student with scholarship/Fellowship

vii. **1099-INT** → If have interest income

viii. **1099-DIV** → If have dividend income

ix. **1099-B** → If have income

x. I-94 (extracted from stage 0)

c. **SYSTEM** → Auto-Classified and extract all data needed for output

d. **ASK** →

2. Simple Validation

- a. Missing Expected Documents
 - b. Unexpected Documents (uploaded document that is not asked)
 - c. False Documents (e.g. no scholarship, but 1098-T shows scholarship)
 - d. Duplicate or unclear uploads
3. Deeper validation
- a. **FICA Recovery**
 Compare the residency verdict from Stage 0 (confirmed NRA, FICA-exempt) against W-2 boxes 4 and 6. If those boxes are nonzero, the employer withheld Social Security and Medicare tax that shouldn't have been taken. Finding card with a dollar figure: "Your employer withheld \$2,678 in Social Security and Medicare tax. As an F-1 student in your first five years, you're exempt from these taxes — you can get this money back. I'll prepare the refund claim (Form 843) as part of your filing." This is the single highest-value finding because it's real money the student didn't know about. Show the exact breakdown: "\$1,860 Social Security (box 4) + \$818 Medicare (box 6) = \$2,678 total."
 - b. **Scholar taxability**
 Compare 1098-T box 5 (total scholarships) against box 1 (tuition and required fees). If box 5 > box 1, compute the taxable excess. Then cross-check: does a 1042-S exist reporting that same taxable amount? Three possible outcomes. First, the numbers match — the 1042-S reports the same taxable scholarship the 1098-T math produces. Finding card: "Your scholarship was \$40,000 and your tuition was \$30,000, so \$10,000 is taxable income. Your school already reported this on your 1042-S and withheld \$1,400 in tax." Second, there's a 1098-T showing a taxable excess but no 1042-S was uploaded. Finding card: "Your scholarship exceeds your tuition by \$10,000 — that portion is taxable. I'd expect a 1042-S from your school reporting this. Check your student portal or contact your school's tax office — it's a different office than the one that sent your 1098-T." Action: upload slot for 1042-S, plus a "I can't find one" option (in which case the system computes taxability from the 1098-T alone). Third, the 1098-T shows no taxable excess (box 5 ≤ box 1). Finding card: "Your scholarship covered tuition only — no taxable scholarship income." Reassuring, and it closes that branch.
 - c. **Treaty Income Reconciliation**
 If a 1042-S exists with a treaty exemption code, verify it makes sense: does the exemption code match the student's country? Is the income code consistent with the income type (scholarship vs wages)? If the student has *both* a W-2 and a 1042-S from the same employer (common when a treaty exempts a portion of wages — the exempt portion goes on the 1042-S, the taxable remainder on the W-2), verify they add up to the total compensation without double-counting or gaps. Finding card: "Your employer paid you \$35,000 total. Under the US-[country] treaty, \$5,000 was exempt — that's on your 1042-S. The remaining \$30,000 is on your W-2. These match up correctly."
 - d. **State Income Crosscheck**
 Compare W-2 box 15 (state) against what's known about the student's situation.

If box 15 shows a state with income tax → the student needs a state return, and the upload checklist should note it. If box 15 shows multiple states (or multiple W-2s from different states) → flag multi-state. If the student is in a no-income-tax state but box 15 shows a *different* state → something unexpected is happening (maybe they interned in California over summer). Finding card: "Your W-2 shows you earned wages in California, which has a state income tax. You'll need to file a California nonresident return in addition to your federal return."

e. Income Completeness

Sum all reported income across all documents and present it as a confirmation. Finding card: "Here's everything I found — \$35,000 in wages (W-2), \$10,000 in taxable scholarship (1042-S), \$47 in bank interest (1099-INT). Total US income: \$45,047. Does this look complete, or did you have other income I'm missing?"

4. Anything else

- a. **ASK** → Do you happen to receive any other tax documents that I have not mentioned? [Yes / No]

Stage 3 - Tax Reduction

1. Treaty Determination

- a. **CONFIRM** → Confirm country based on Passport (Extracted from stage 1)
- b. **SYSTEM** → Treaty look up for the country.
 - i. Does a treaty exist?
 - ii. Which student/trainee articles apply (exempt amount, income type, time limit, etc.)
 - iii. Determine form 8833

2. Deduction

- a. **ASK** → Have you done any charitable contribution? (for itemized deduction)
- b. **SYSTEM** → If country = india → Standard deduction may apply, everyone else itemized deduction. India compares Standard vs Itemized deduction, other just use itemized if available.

Stage 4 - Output (the brain & calculation) - Start from Paper Path for now

SYSTEM → Put all of those into frameworks and output the documents in PDF (for now)

- Federal Tax Return (With Income)

- 1. 1040-NR (Always)
- 2. Schedule OI (Always)
- 3. Form 8843 (Always)
- 4. Schedule NEC (if passive income)
- 5. Schedule A (if itemizing)
- 6. Form 8833 (if treaty not in 1042-S. If india treaty)
- 7. Form 843 + Form 8316 (If FICA withheld)
- 8. W-2 Copy B (If working)
- 9. 1042-S (if received one)

- FICA Recovery Claim

- **ASK → Make them an email to their employer to get a refund from their employer.**
 - **If employer don't want to help, then file this.**
 1. Form 843 (Refund Claimed)
 2. Form 8316 (Supporting Statement)
 3. W-2
 4. Passport
 5. I-94
 6. I-20
 7. EAD Card (if OPT)
- **No Income**
 1. **Form 8843**

SYSTEM → Build Summary, Explanation, and Instruction